ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 1 [10 marks]

Q1(a) [4 marks]

Pixel pitch = 0.005 mm, sensor width = 1280x0.005 = 6.4 mm. how to calculate the belt field of view.

Q1(b) [3 marks]

The smallest resolvable force is 0.122 N.

Q1(c) [3 marks]

$$F = e^(AT) = e^(-0.2) = 0.818731$$

Question 3 [10 marks]

Q3(a) [2 marks]

The estimate is noisy and can fluctuate from sample to sample.

Q3(b) [2 marks]

Q3(c) [4 marks]

A-LC=[[-ℓ_1,1], [-ℓ_2,-2]]. The desired polynomial is $(s+10)^2 = s^2+20s+100$.

Q3(d) [2 marks]

An observer can estimate velocity from position measurements. A filter can be safer because it smooths noise.

Question 4 [10 marks]

Q4(a) [2 marks]

Closed-loop characteristic polynomial is $s^2+(a+bk_2)s+(c+bk_1)=s^2+4s+8$, giving poles -2±2j. At DC, θ/r = bNff/(c+bk₁), so unity gain requires Nff=(c+bk₁)/b=(6+2)/1=8.

Comparing coefficients with $s^2 + 4s + 8$ gives $a + bk_2 = 4$ and $c + bk_1 = 8$. The feedforward gain is chosen from the nominal plant so the reference scaling is unity; here this gives $N_{ff} = 8$. The closed-loop roots of $s^2 + 4s + 8$ are $-2 \pm 2j$.

Q4(b) **[2 marks]**

Q4(c) **[4 marks]**

Q4(d) **[2 marks]**

Integral action helps eliminate error. A problem is that the controller can become unstable.